

CONIFERS

Conifers evolved over 300 million years ago, long before the world’s first broadleaved trees. With their tough waxy leaves, conifers thrive in harsh climates. They are less diverse than broadleaved trees, but they dominate forests on cold mountains, and in the far north.

DIVISION	PINOPHYTA
CLASS	1
ORDER	2
FAMILIES	6
SPECIES	About 600



Between areas of open tundra, conifers grow far to the north of the Arctic Circle, forming the largest forest on Earth.

Although relatively few in species, conifers include the world’s tallest trees, as well as the heaviest, the longest-lived, and some of the most widespread. Conifers are traditionally placed in a group known as gymnosperms, along with cycads, ginkgos, and gnetophytes. Unlike broadleaved trees, they do not have flowers and produce pollen and seeds in cones.

LEAVES AND CONES

Most conifers are evergreen, with highly resinous leaves that are good at withstanding cold winds and strong sunshine. Pines have slender needles, arranged singly or in bundles, while other conifers often have linear leaves or flat scales. A few conifers – including larches and some redwoods – are deciduous, with soft leaves that are shed each year.

Conifers have two types of cone, which normally grow on the same tree. Male cones produce pollen. Small, soft, and often very numerous, they typically appear in spring, and

wither soon after shedding their pollen into the air. The larger female cones contain one or more seeds. They can take several years to develop, and usually become hard and woody when mature.

The cones of some species, including firs and cedars, release their seeds by slowly disintegrating on the tree. Pine cones remain intact, and often remain on the tree long after they are ripe. Most of them shed their seeds in dry weather, when their scales open wide, but some hold onto their seeds until they are scorched. This is a specialized adaptation that allows pines to recolonize areas that have been burned by forest fires.

A few conifers, the yews, junipers, and podocarps, have small, berry-like cones with fleshy scales that do not fully enclose the seed. These are scattered by birds as they feed.

✓ CONIFERS IN CONTROL

Conifers often form extensive single-species stands. These pines are growing in California’s Yosemite National Park in the USA.

